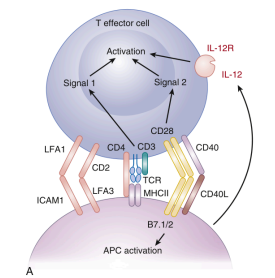
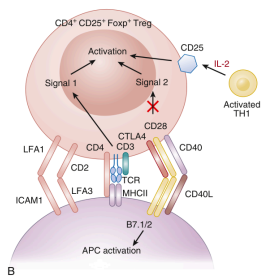


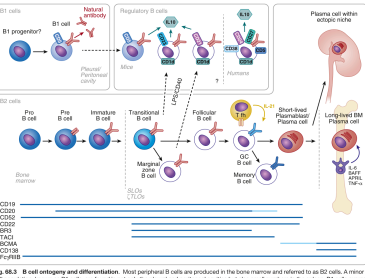
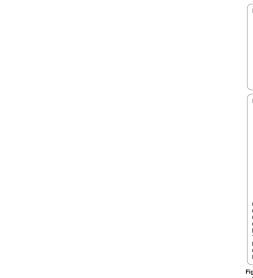
Fig_68_1



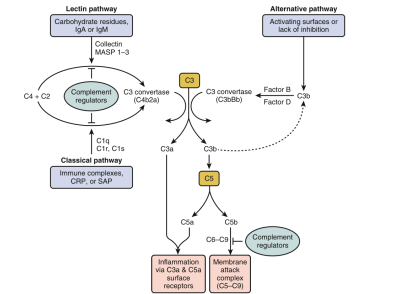
Fig_68_2



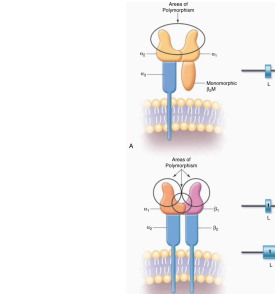
Fig_68_3



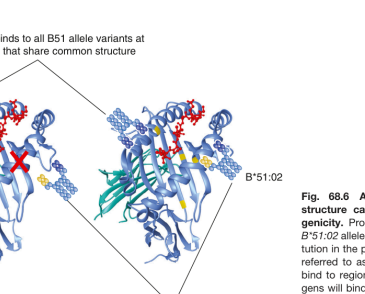
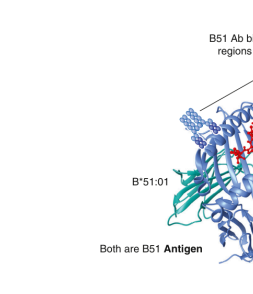
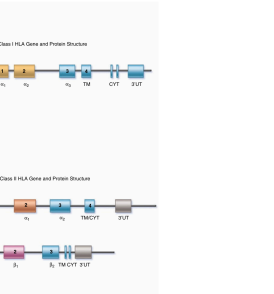
Fig_68_6



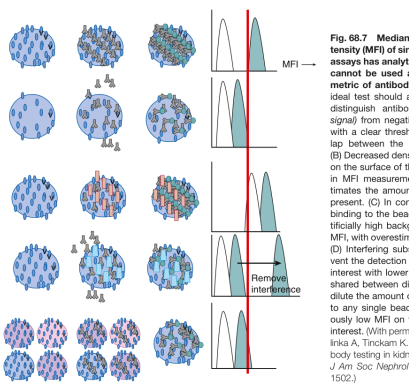
Fig_68_7



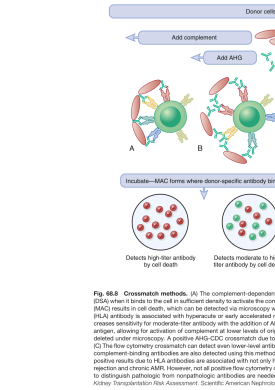
Fig_68_8



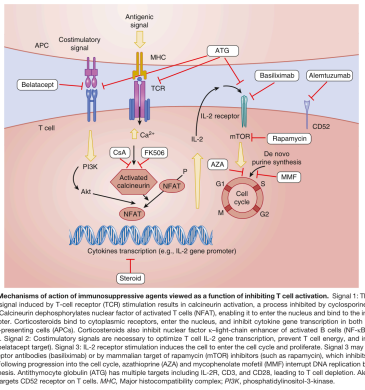
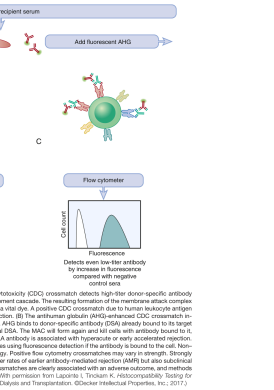
Fig_68_12



Fig_68_14



Fig_68_16



Fig_68_20

Table 68.1 Common Human Leukocyte Antigen Genes with Multiple Human Leukocyte Antigens Associated

Human Leukocyte Antigen Gene	Antigens Encoded By Gene
HLA-B*14	B64 or B65
HLA-B*15	B62, B63, B71, B72, B75, B76, or B77
HLA-B*40	B60 or B61
HLA-C*03	Cw9 or Cw10
HLA-DRB1*03	DR17 or DR18
HLA-DQB1*03	DQ, DQ8, or DQ9

Table_68_1

Table 68.2 Human Leukocyte Antigens Mismatching Considered From The Perspective of The Recipient Immune System

Human Leukocyte Antigen Locus	A*	B*	Cw*	DRB1*	DQA1*	DQB1*	DPA1*	DPB1*	Total Mismatches
Recipient	11	15(62)	15(75)	04	12	06:01	03:01(7)	01:03	02:02
	24		08	14	–	–	02	21:01	
Donor	2	46	01	09	03	03:03(9)	02-	05:01	
	–	52	14	–	–	–	–	09:01	
Mismatches	1	2	2	1	1	1	0	2	10
Recipient	2	46	01	09	03	03:03(9)	02-	05:01	
	–	52	14	–	–	–	–	09:01	
Donor	11	15(62)	15(75)	04	12	06:01	03:01(7)	01:03	02:02
	24		08	14	–	–	02	21:01	
Mismatches	2	2	2	2	1	1	1	2	13

Molecularly determined alleles at each human leukocyte antigen (HLA) locus in recipient and donor in two different scenarios (upper and lower section of table), showing the number of mismatches. In the case where either a recipient or donor is homozygous for 1 or more antigens, there is a difference in mismatches depending on which direction the mismatches are considered. In the first example, considering the foreign HLA donor, antigen burden yields 10 mismatches in the host versus graft direction. However, if the donor and recipient had the opposing HLA typing, it would give a total of 13 mismatches from the perspective of this recipient.

Table_68_2

Table 68.3 Quantifying Human Leukocyte Antigens Versus Eplet Mismatches

Human Leukocyte Antigen Locus	A*	B*	Cw*	DRB1*	DQA1*	DQB1*	Total Mismatches
Recipient	01:01	07:02	07:01	03:01	01:03	02:01	
	03:01	08:01	07:02	13:09	05:01	06:03	
Donor	02:01	44:02	03:01	01:01	01:02	06~2	
	11:01	54:01	05:01	04:05	03:01	03:02	
HLA antigen mismatches		6			4		10
Eplet mismatches		35			51		86
Recipient	01:01	51:01	07:01	03:02(18)	03:01	02:02	
	24:02	35:01	01:02	07:01	04:01	04:02	
Donor	23:01	52:01	15:02	03:01(17)	05:01	02:01	
	36:01	53:01	18:01	11:04	–	03:01	
HLA antigen mismatches		6			4		10
Eplet mismatches		4			23		27

HLA antigen mismatch is only a crude estimate of the degree of mismatch between donors and recipients. In these examples, despite the same number of class I and class II antigen mismatches, the first pair has almost 3 times the number of eplet mismatches (not shown), representing a more potent alloimmune stimulus.

Table_68_3

Table 68.4 Clinical Histocompatibility Testing Menu and Associated Interpretations and Actions

Timepoint	Clinical Question(S) Applicable	Tests Used	Interpretation and Potential Actions
Pretransplant	How sensitized is the patient? How difficult is it to find a (HLA-based) suitable donor?	HLA typing HLA AB screening and identification	If reduced access to donors from high cPRA, then access to more donors is optimal (increased sharing for acceptable mismatches, kidney paired donation programs) or desensitization to reduce current burden of circulating antibody.
	What is the patient's risk of a memory response post transplant?	HLA typing HLA AB screening in the context of a good clinical history of sensitizing events	All patients with HLA antibodies, as well as those with sensitizing events, should be considered at higher risk of memory responses post transplant; closer follow-up post transplant may be warranted.
At transplant	Is the patient PREDICTED to have HLA antibodies to the donor?	The virtual crossmatch: donor typing PLUS	Positive HISTORICAL VXM: Patient is at higher risk of memory response. Some programs do not transplant; others augment immunotherapy or monitoring.
	Will the cell-based crossmatch before transplant be positive?	HLA AB identification (current and historical results)	Positive CURRENT VXM: Patient is at higher risk of AMR. Transplant decision making depends on strength of antibody (AB test plus cell-based crossmatch).
Post transplant	Has the patient developed de novo HLA antibodies? (no graft dysfunction)	Donor typing from transplant Post-transplant AB testing	If positive, may indicate risk of future or current subclinical AMR? Guides further diagnostics or monitoring, noting certain HLA loci portend a worse prognosis (class II).
	Does the patient have AMR? (with or without graft dysfunction)	Donor typing Post-transplant AB testing In context of biopsy if available	HLA antibodies are supportive of a diagnosis of AMR but may not always be present.

AB, Antibody; AMR, antibody-mediated rejection; HLA, human leukocyte antigen; VXM, virtual crossmatch.

Table_68_4